UNIT 5 · TOPIC 6(B)

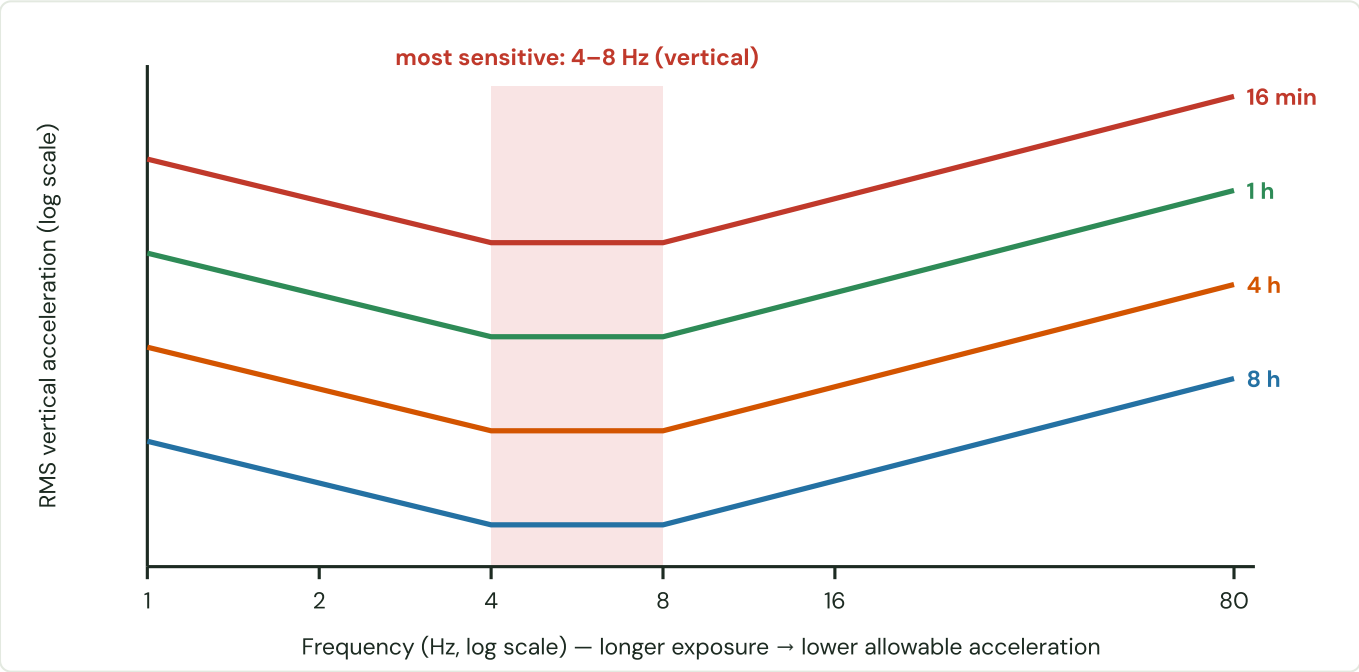
Vibration — Whole-Body, Seat Suspension and Hand-Arm

Syllabus topic: Effect of noise and vibration on work performance

Whole-body vibration (WBV) and work performance

- Ride vibration intensity rises with ground speed and often becomes intolerable.
- **Matthews (1972):** most large tractors ran below **two-thirds** of maximum power because operators could not withstand the ride at full speed (p. 210) — vibration directly limits productivity.
- **Rossegger & Rossegger (1960): 71.3%** of tractor drivers showed pathological spine deformations — second only to truck drivers (**80%**) among the occupations studied (Table 9-3, p. 212). Gastrointestinal complaints were also elevated (Table 9-4).

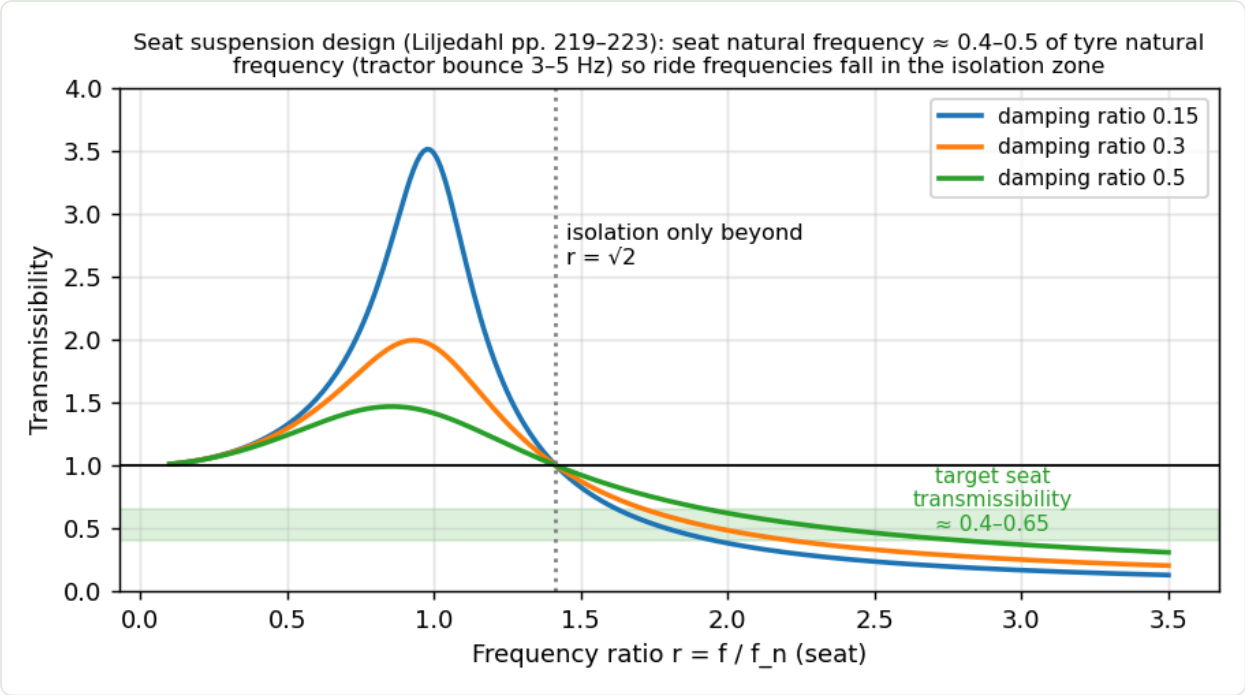
Human tolerance — ISO 2631



ISO 2631 fatigue-decreased-proficiency boundaries — concept sketch after Liljedahl Fig. 9-6 (redrawn)

- **ISO 2631** (Guide for Evaluation of Human Exposure to Whole-Body Vibration) sets **fatigue-decreased-proficiency (FDP)** boundaries of RMS acceleration vs frequency and exposure time (Fig. 9-6).
- The seated human is MOST SENSITIVE to VERTICAL acceleration at **4–8 Hz** (body-organ resonance) (pp. ~216–218).
- Unsprung wheel tractors bounce at an undamped natural frequency of **3–5 Hz** (p. 213) — squarely in the sensitive band. Hence seat suspension matters.

Seat suspension design (pp. 219–226)



Seat transmissibility vs frequency ratio — isolation only beyond r = √2

TRANSMISSIBILITY = output ÷ input vibration intensity

- Seat suspension = spring + damper (+ mass).
- Design target: transmissibility **0.4–0.65** — the seat passes less than two-thirds of the chassis vibration (pp. 219, 223).
- Seat undamped natural frequency ≈ **0.4–0.5** of the tyres' natural frequency, so working frequencies land beyond **r = √2**, in the isolation zone (p. 223).
- Elements: variable-rate springs, shock absorbers, torsion-bar or pneumatic static-deflection compensation; adjust for operator WEIGHT and height.
- Evaluation uses ISO 2631 a_z criteria (p. ~222). More seat-suspension targets: Topic 10a.

Two transmissibility figures appear in the source notes: 0.4–0.65 (satisfactory range, pp. 219, 223 — this topic) and 0.4–0.5 (passive-suspension design target, pp. 223–224 — Topic 10a). Read the MCQ options against the context.

Hand-arm vibration (HAV)

- **Sources:** power-tiller handlebars, brush cutters, chainsaws, sprayers.
- **Effects:** vibration white finger (Raynaud's), numbness, reduced grip.
- **Assessment:** ISO 5349.
- **Controls:** damped/isolated handles, anti-vibration gloves, limited daily trigger time, engine balancing. Power-tiller handle shake is a recognised Indian problem.

Practice MCQs

Q50. Ride vibration intensity generally increases with:

(a) Lower tyre pressure
(b) Ground speed
(c) Engine oil grade
(d) Paint

Q51. Matthews (1972) found most large tractors operated below 2/3 maximum power because:

(a) Weak engines
(b) Operators could not withstand the ride at full speed
(c) Fuel cost
(d) Law

Q52. Rossegger's X-ray study found spine deformations in what share of tractor drivers?

(a) 25%
(b) 43%
(c) 71.3%
(d) 95%

Q53. The only occupation worse than tractor driving for spines in that study:

(a) Miners
(b) Truck drivers
(c) Bus drivers
(d) Builders

Q54. The international guide for human whole-body vibration exposure is:

(a) ISO 500
(b) ISO 2631
(c) ISO 730
(d) ISO 9001

Q55. The seated human is most sensitive to vertical vibration at:

(a) 0.1–1 Hz
(b) 4–8 Hz
(c) 50–100 Hz
(d) 1000 Hz

Q56. Undamped bounce natural frequency of wheel tractors is about:

(a) 0.5–1 Hz
(b) 3–5 Hz
(c) 15–20 Hz
(d) 50 Hz

Q57. ISO 2631 'FDP' boundaries stand for:

(a) Full design power
(b) Fatigue-decreased proficiency
(c) Field data points
(d) Free damping period

Q58. Seat transmissibility is defined as:

(a) Input/output vibration
(b) Output/input vibration intensity
(c) Spring rate
(d) Damping ratio

Q59. Satisfactory seat transmissibility is about:

(a) 1.5–2
(b) 1.0
(c) 0.4–0.65
(d) 0.05

Q60. The seat's natural frequency should be what fraction of the tyres' natural frequency?

(a) 2×
(b) Equal
(c) 0.4–0.5
(d) 0.1

Q61. Vibration isolation occurs only when the frequency ratio exceeds:

(a) 0.5
(b) 1
(c) $\sqrt{2}$
(d) 3

Q62. Seat suspensions include which elements?

(a) Variable-rate springs
(b) Shock absorbers
(c) Weight adjustment for the operator
(d) All of these

Q63. Hand-arm vibration exposure is assessed per:

(a) ISO 5349
(b) ISO 11783
(c) IS 9020
(d) SAE J833

Q64. 'Vibration white finger' results from:

(a) Whole-body vibration
(b) Prolonged hand-transmitted vibration
(c) Cold water
(d) Noise

Q65. A well-known Indian HAV source is:

(a) Seed drill
(b) Power tiller handlebars
(c) MB plough
(d) Level

Answers

Q50 — (b) Ground speed
Q51 — (b) Operators could not withstand the ride at full speed — *pp. 210 — Matthews' two-thirds-power finding.*
Q52 — (c) 71.3% — *Rossegger, Table 9-3, p. 212 — 71.3%.*
Q53 — (b) Truck drivers — *Truck drivers 80%.*
Q54 — (b) ISO 2631
Q55 — (b) 4–8 Hz
Q56 — (b) 3–5 Hz — *p. 213 — tractor bounce 3–5 Hz.*
Q57 — (b) Fatigue-decreased proficiency
Q58 — (b) Output/input vibration intensity
Q59 — (c) 0.4–0.65 — *pp. 219, 223 — transmissibility target.*
Q60 — (c) 0.4–0.5 — *p. 223 — seat-to-tyre frequency ratio.*
Q61 — (c) √2
Q62 — (d) All of these
Q63 — (a) ISO 5349
Q64 — (b) Prolonged hand-transmitted vibration
Q65 — (b) Power tiller handlebars

Sources: Liljedahl et al. — Tractors and their Power Units, Ch. 9 (noise pp. 207–211; vibration pp. 210–218; seating pp. 219–226); ISO 2631; OSHA criteria; CIAE Bhopal improved-tool studies; NPTEL ergonomics lectures.